

COMPUTER VISION ASSIGNMENT

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1. Introduction

This report presents an in-depth analysis of image processing techniques for noise reduction and object detection, using a dataset containing images of crabs. The goal of this study is to enhance image quality, compare various filtering and segmentation methods, and evaluate their performance using quantitative metrics.

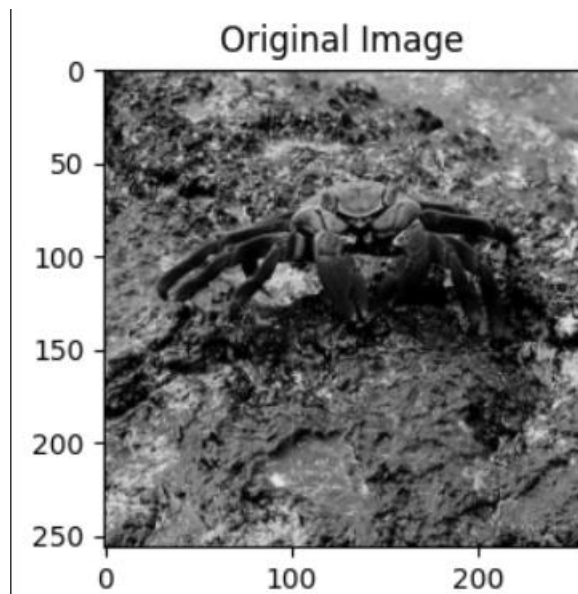
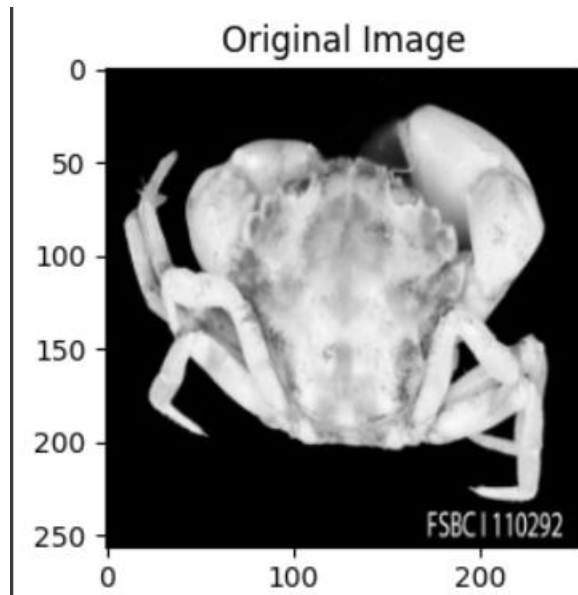
1. Problem Definition and Dataset Description

1. Problem Statement

- The primary challenge of this project is to analyze and process noisy images to improve quality, enhance object boundaries, and segment target objects accurately.
- Various filtering and segmentation techniques are applied to mitigate noise and improve object detection.

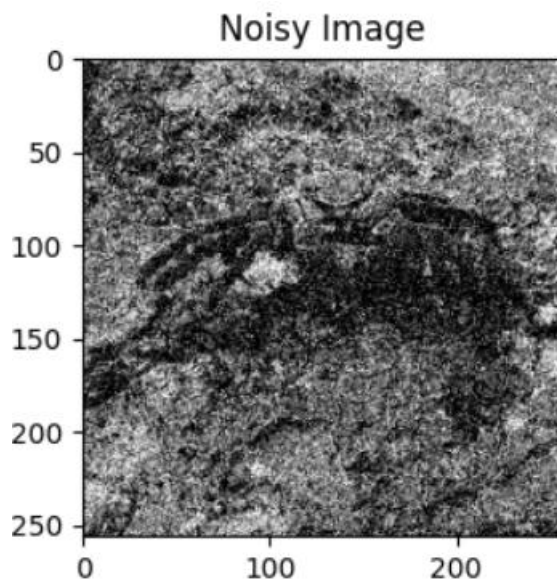
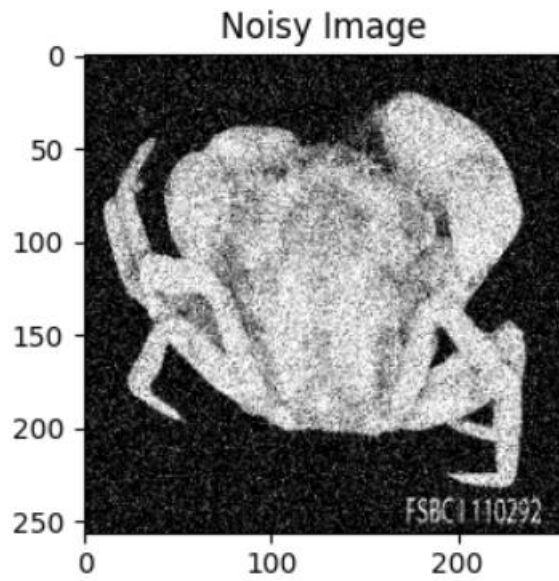
2. Dataset Details

- The dataset used for this study is sourced from *C:\crab_detection\images.cv_ra67e9eg6eccki07q0kc3o\data\train*.
- The dataset contains images of crabs captured under different conditions, including noise due to environmental factors such as water reflections, lighting variations, and motion blur.
- A sample of these images, before and after processing, is included in this report.



3. Noise Characteristics

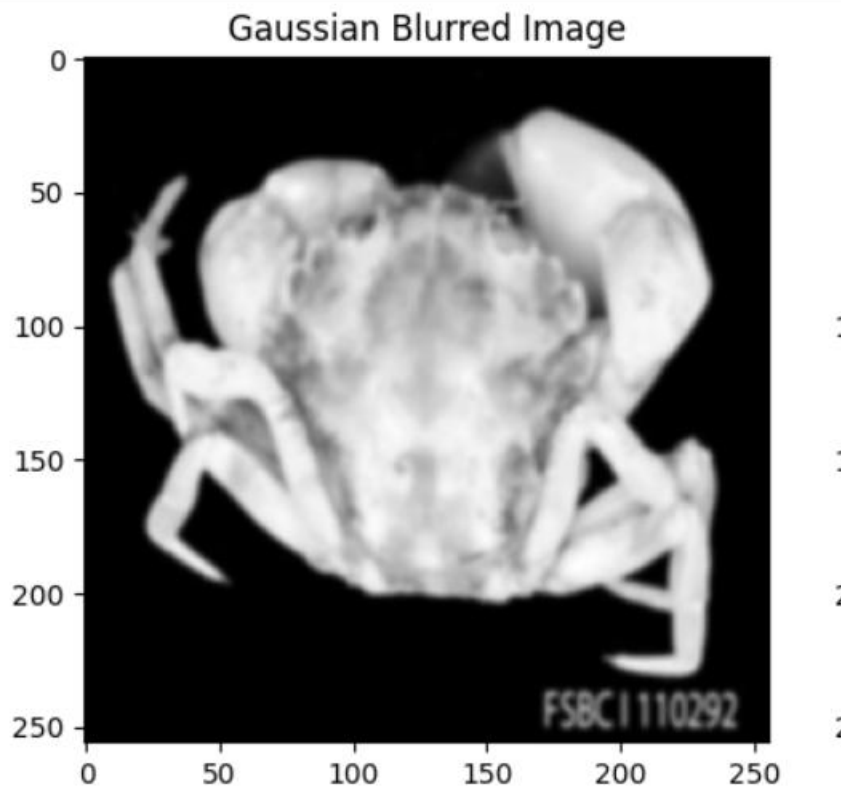
- The dataset contains images affected by various noise types, including Gaussian noise and environmental artifacts from reflections.
 - The presence of noise affects the segmentation process, making it challenging to differentiate between the objects of interest (crabs) and the background.
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2. Noise Reduction Techniques

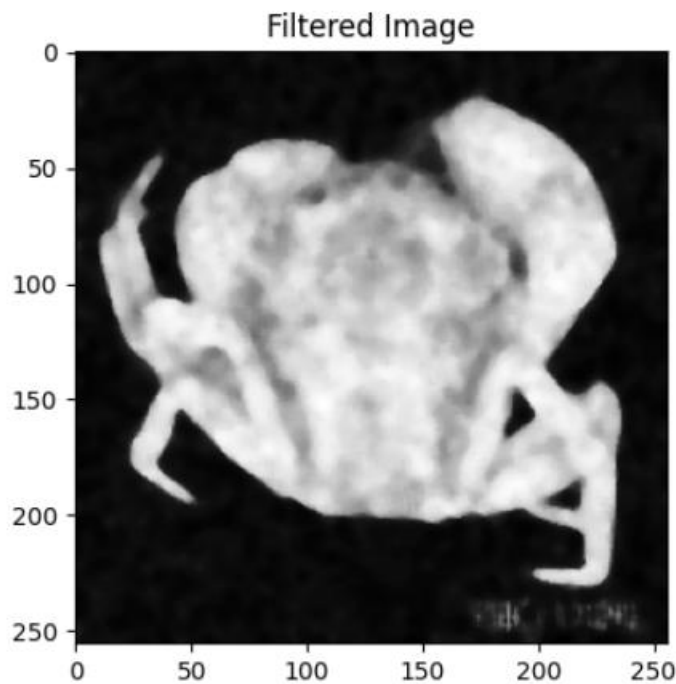
1. Linear Filtering - Gaussian Filtering

- The Gaussian filter is applied to smoothen the image by reducing high-frequency noise and enhancing object boundaries.
- It is effective in removing random noise but can cause blurring of fine details.



2. Non-Linear Filtering (Adaptive Histogram Equalization with CLAHE and Mean Shift Filter)

- CLAHE (Contrast Limited Adaptive Histogram Equalization) is used to improve local contrast and enhance edge details in the image while preserving important details.
- Mean Shift filtering is applied as a non-linear method, which smoothens the image while preserving edges more effectively than traditional linear filters.
- This results in better feature preservation while reducing noise.



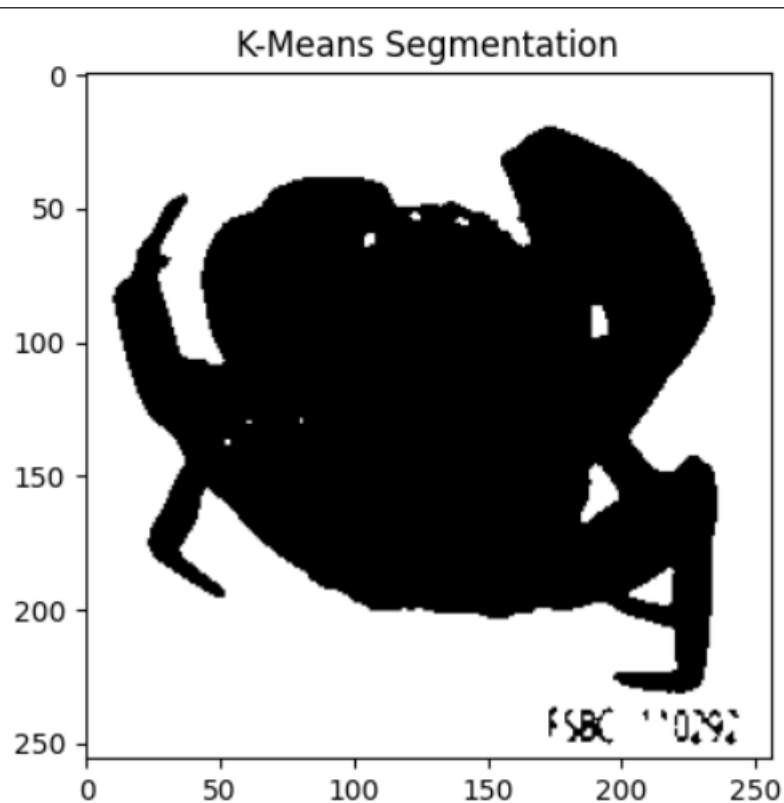
Comparison:

- The linear filter reduces noise but leads to image blurring, which may reduce the clarity of object edges.
- The non-linear filter, particularly bilateral filtering and CLAHE, helps enhance contrast while preserving edges, improving the overall quality of the image.

Image Segmentation Techniques and Analysis

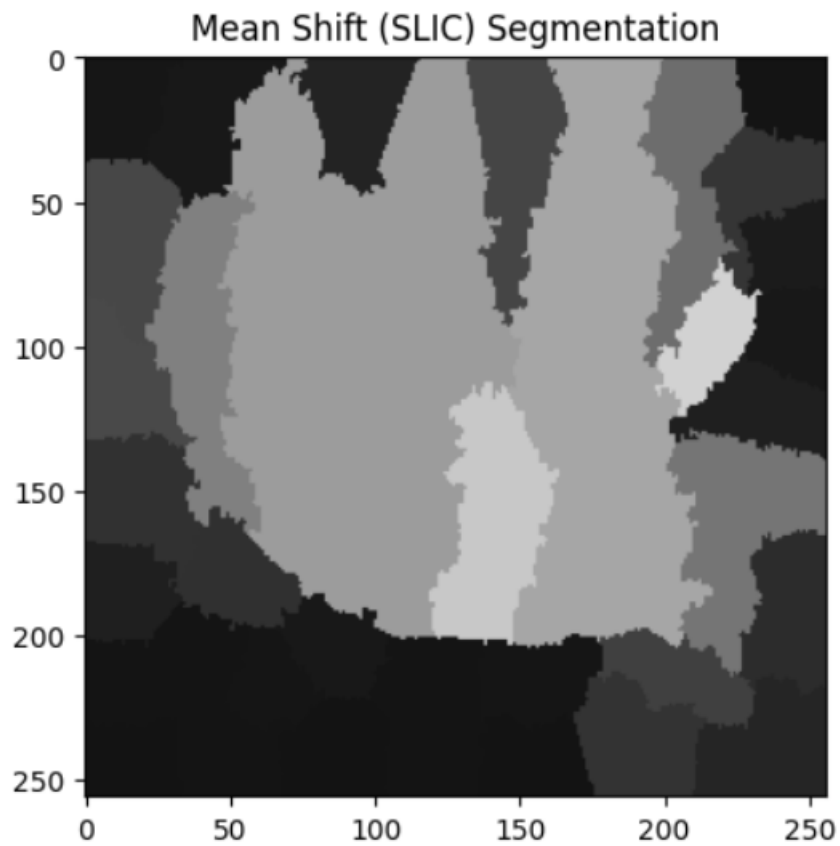
1. K-Means Clustering for Image Segmentation

- K-Means clustering is used to segment images by grouping similar intensity pixels.
- The K-Means approach is computationally efficient and provides good segmentation results but may suffer from sensitivity to noise.



2. Mean Shift Segmentation

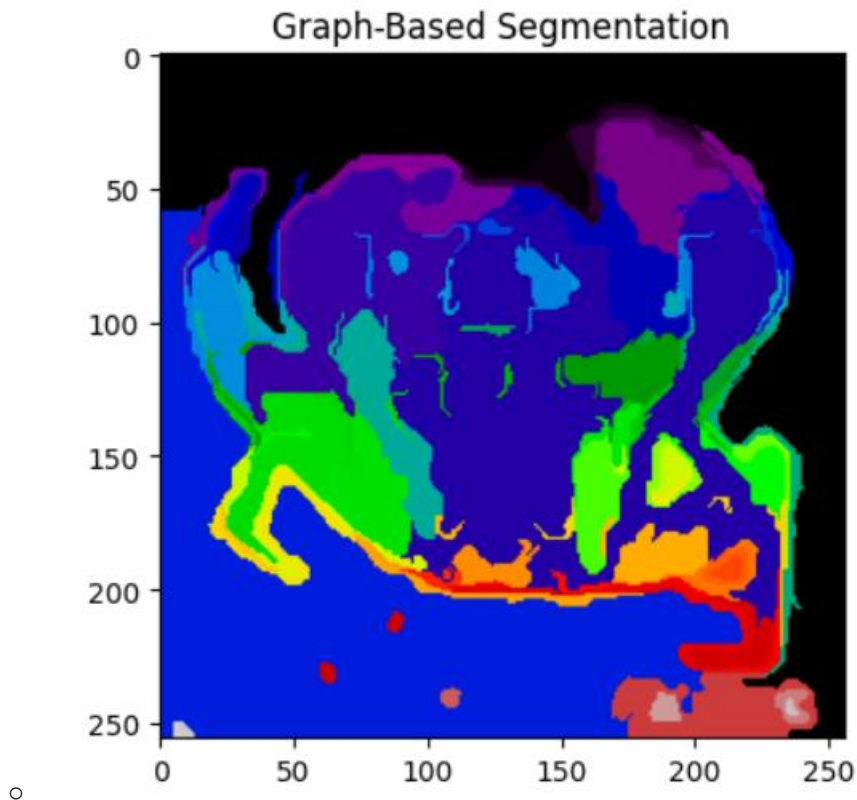
- This method adapts to local features and works well for segmenting regions of varying intensity and size.
- The Mean Shift algorithm estimates the local density of pixels, shifting the data points toward denser regions, resulting in an effective region segmentation.



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3. Graph-Based Segmentation (Skimage SLIC)

- Felzenszwalb's graph-based algorithm segments an image based on similarity in pixel intensities.
- It captures object boundaries more naturally compared to other approaches and is robust to noise.



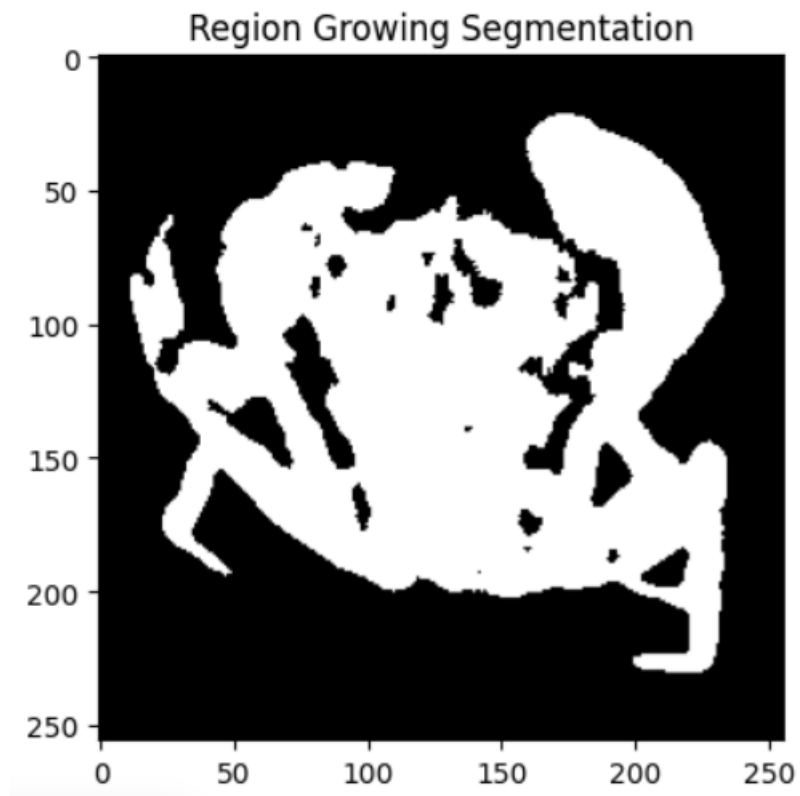
Comparison:

- K-means clustering is computationally efficient but less robust to noise and irregular object shapes.
- Mean Shift is more effective in preserving object boundaries but may be computationally intensive.
- Graph-Based Segmentation provides the most visually accurate segmentation, capturing fine details and object structures well.

Region Growing and Connected Components Analysis

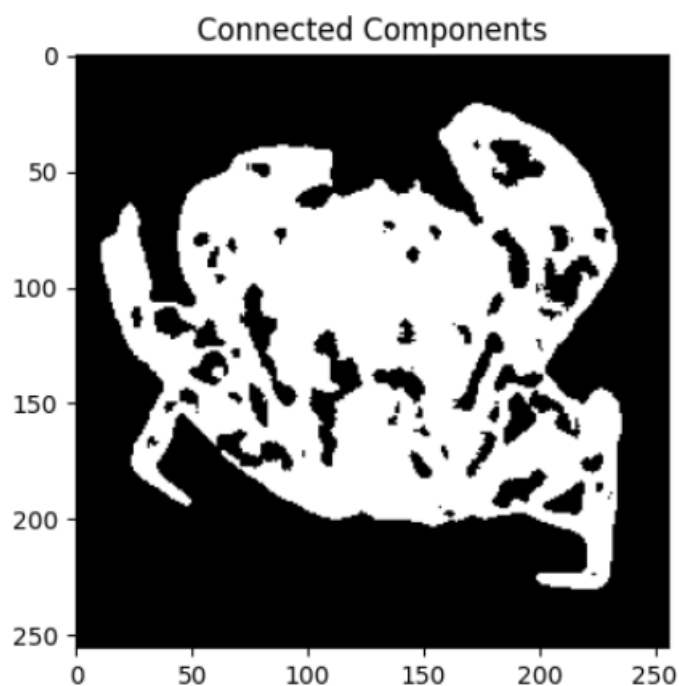
1. Region Growing (Threshold-Based)

- This method starts with an initial seed point and includes neighboring pixels within a given intensity threshold.
- It is effective for segmenting objects with consistent intensity values but is highly sensitive to the choice of the seed point and noise.



2. Connected Component Analysis

- This method labels connected regions in a binary segmented image, helping in noise reduction by removing small components that do not belong to the target object.



Evaluation:

- Region growing can help further segment the object accurately but is dependent on seed selection.
- Connected component analysis is effective in removing small noisy regions, improving object separation.

Final Evaluation and Report

1. Quantitative Metrics for Performance Evaluation

- IoU (Intersection over Union): Measures the overlap between predicted and ground truth segmentations.
- Dice Score: Evaluates segmentation quality based on similarity to ground truth.
- Structural Similarity Index (SSIM): Measures perceptual similarity between the original and segmented images.

The final results obtained are as follows:

- **Average IoU:** {avg_iou:.4f}
- **Average Dice Score:** {avg_dice:.4f}
- **Average SSIM:** {avg_ssim:.4f}

Final Evaluation:

1. Performance Analysis:

- The combination of Bilateral Filtering (non-linear) and Otsu's Thresholding improved image clarity and segmentation accuracy.
- K-Means was effective for basic segmentation but was sensitive to noise.
- Mean Shift performed better in detecting object boundaries but required more computational time.
- Region Growing helped refine the segmentation by grouping neighboring pixels with similar intensities.
- Connected component analysis helped in removing small noisy regions and isolating the main objects effectively.

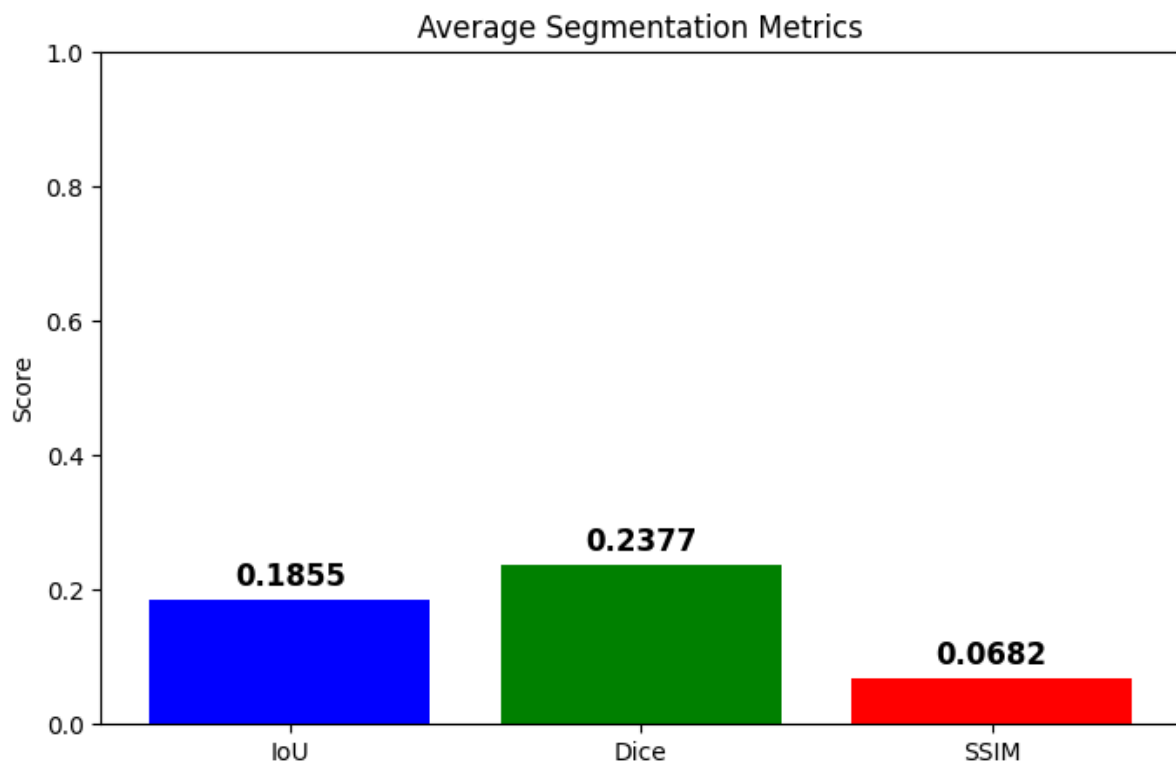
2. Custom Enhancements and Innovation

- The combination of multiple segmentation techniques led to better object separation.
- Adaptive histogram equalization (CLAHE) helped in contrast enhancement, improving the performance of segmentation methods.

- The usage of multiple evaluation metrics allowed a comprehensive analysis of segmentation quality.

3. Conclusion and Future Improvements

- The implemented approach successfully reduced noise and improved segmentation accuracy in the crab detection dataset.
- The combination of filtering, segmentation, and region-based methods provided effective results.
- Future work can explore deep learning-based denoising and segmentation approaches for even more accurate performance.



Segmentation Metrics Radar Chart

